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# Honeybee Swarm Dynamics: Investigating the Relationship Between Individual Decision Making and Collective Foraging

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## Abstract

Honeybees are social insects that utilize pheromone signals to direct each other, resulting in emergent swarm behavior. Research has shown that bees can solve the shortest path problem and locate their queen by directing these pheromone signals, or “scenting,” to create a communication network collectively; however, little work has been done to study how the behavior of the bees changes when in the presence of the swarm. This study aims to investigate how the decisions of individual honeybees affect the foraging and aggregation process of the collective. We utilize state-of-the-art video object tracking and segmentation tools to gather time-series data on the scenting behaviors of individual bees when separated from their queen. Combining this with additional computer vision approaches described in the literature, we analyze the behavior patterns of the individuals as the swarm forages for and aggregates around the queen. Through this analysis, we anticipate the time-series data from individual honeybees will display a predictable relationship between scenting and exploring that allows for the emergent pathfinding capability of the swarm. This improved tracking of individual honeybees will allow us to increase the accuracy of the multi-agent reinforcement learning model to predict the behaviors of the swarm. These improvements directly impact swarm robotics and tasks such as patrolling, disaster recovery, and search and rescue, where it is important to create a low-resource distributed system that can rapidly adapt to new operating conditions.

## 1. Introduction

A vast body of research has surrounded the topic of honeybee swarms. To become a coherent swarm, bees locate their queen by tracking her pheromones. Previous work has shown that bees collectively create a scenting-mediated communication network by arranging in a specific spatial distribution where there is a characteristic distance between

individuals and directional signaling away from the queen. Rather than depositing static information in the environment, individual bees locally sense and globally manipulate the physical fields of chemical concentration and airflow. Individual bees act as receivers and senders of signals by using the Nasonov scenting behavior, releasing pheromones from the glands and fanning their wings to direct the signals backward. In this network, scenting bees stand at a characteristic distance from their neighbors while dispersing signals, which suggests a concentration threshold in the activation mechanism of individual bees’ scenting behavior. The scenting events are highly correlated with the collective aggregation around the queen (Nguyen, et al. 2021).

Despite previous research showing the correlation between scenting events and collective aggregation around the queen, little work has been done to study how the behavior of individual bees changes when in the presence of the swarm. This was because we were only able to detect the states of the bees; however, previous methods were not able to track them over longer periods of time.

## 2. Methods

To track individual bees and understand their behavior in the presence of the swarm, we established an environment in which worker bees search for a stationary caged queen in a semi-two-dimensional (2D) arena. We recorded the search and aggregation behavior of the bees from an aerial view with tests containing one, two, three, and four worker bees. To extract data from the recordings, we then developed a markerless, semi-automatic, highly accurate tracking and segmentation algorithm using a variety of computer vision methods. This pipeline improves upon previous work by allowing us to detect and track individual bees throughout the video, while also identifying the positions and orientations of scenting bees each frame.

### 2.1. The Algorithm

`get_background(video)` - Since the background of our video is stationary we utilize a standard method of video background subtraction by defining the background to be the median of each frame. By using this temporal median

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**Algorithm 1** Honeybee Tracking and Segmentation

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background = get_background(video)
for frame  $\in$  video do
    processed = preprocess(frame, background)
    contours = form_contours(processed)
    tracks = create_tracks(contours, prev_tracks)
    split_tracks = split_groups(tracks)
    frame = draw_tracks(frame, split_tracks)
    show_frame(frame)
    prev_tracks = split_tracks
end for

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approach, we can approximate the background pixels by creating an image that contains the

preprocess(frame, background) - To preprocess our frame, we utilize traditional image processing techniques to isolate the honeybees. We start by subtracting the background from the current frame. From there, we blur the resulting image in order to reduce noise for our next step, thresholding. We apply a binary threshold to the grayscale image to isolate the bees. After this step, we apply a variety of morphological image operations in order to reduce the noise and improve the consistency of detecting the honeybees within the thresholded image. As a final step, depending on the quality of the resulting video we remove pixels of irrelevant artifacts such as the queen's cage or the border of the arena.

form\_contours(processed) - We utilize the method described by Suzuki et al. (1985) to form contours from the topological features of the preprocessed frame. We then leverage the topological structural analysis of the method to lift the bee contours from the background. Finally, an area filter is applied to remove noise from the resulting contours.

create\_tracks(contours, prev\_tracks) - A "track" is an object that contains labels, a center point, and a contour. We form these track objects and assign labels based on a greedy search algorithm that identifies the nearest previous track and takes that label.

split\_groups(tracks) - This function is the bulk of the bee tracking pipeline. In order to segment the individual bees from the group, we leverage the watershed algorithm. The watershed transformation treats the image it operates upon like a topographic map, with the brightness of each point representing its height, and finds the lines that run along the tops of ridges. Specifically we implement one of the variants of watershed, non-parametric marker-based segmentation algorithm, described by Meyer (1992). Since the watershed algorithm requires landmark points to start, we utilize the previous center points for each of the track objects. If the current frame is the first frame and group separation is needed, then we enter a manual landmark selection mode where users can select the centers of the honeybees then run

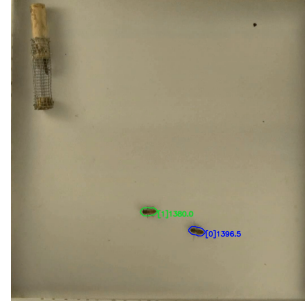
the watershed algorithm. After separating the honeybees, we then use the same greedy algorithm to relabel the contours and form the track objects.

Despite this process functioning a majority of the time, there are edge cases where it can fail such as when bees are on top of one another or if they rotate while being grouped. In order to mitigate these issues, we implemented a progressive color thresholding technique where bee contours are recalculated based on thresholding based on the color at the watershed landmark. We start with a large range of color values and progressively restrict the range until the watershed algorithm can separate the contours.

This additional step fixes a myriad of edge cases but occasionally the algorithm still cannot separate the contours. In this scenario, we enter a manual mode where users can select the landmarks for the watershed algorithm. Including this manual mode ensures that the resulting tracking objects are highly accurate and temporally consistent.

draw\_tracks(split\_tracks) - Simply visualizes the contours of the individual bees and their respective labels.

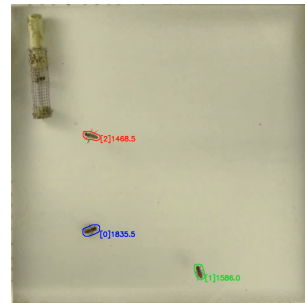
## 2.2. Figures



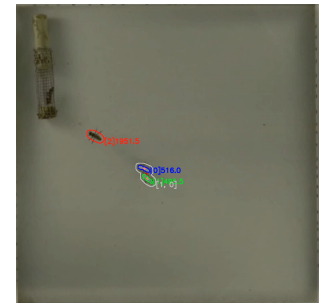
(a) Two honeybees detected by the tracker



(b) Two clustered honeybees detected by the tracker



(c) Three honeybees detected by the tracker

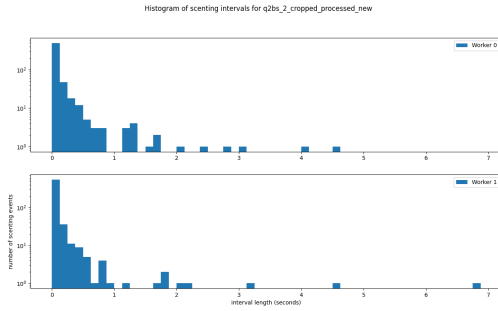


(d) Three clustered honeybees detected by the tracker

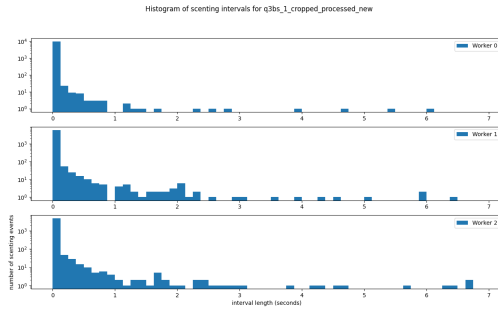
### 3. Results

“**intervals\_histogram**” plots the distribution of scenting events with the specified interval length, where the x-axis is the length (in seconds) of the interval between scenting events and the y-axis is the number of scenting events that had that interval length.

An interval is defined as the length of time between the current scenting event and the previous one. Along with this, we decided to group intervals by the nearest 1/8th of a second so the plot would be easily interpretable (note that the y-axis is on a logarithmic scale because the shorter intervals tend to dominate the graph).



(e) Two honeybees intervals\_histogram



(f) Three honeybees intervals\_histogram

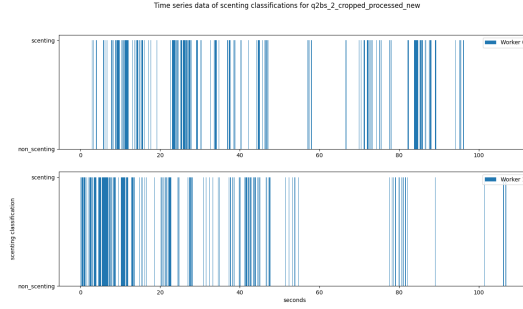
“**time\_series\_bar**” plots the same data as “time\_series\_scatter” however each bar represents a frame (1/30th of a second) and the height of the bar is 1 if the bee was scenting during the given frame and 0 if not.

Notice how there are significantly less scenting events for the test with two honeybees when compared to three.

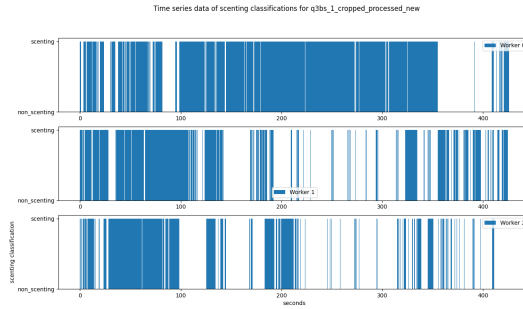
It is important to point out that the test with three honeybees had much more consistent scenting events when compared to the test with simply two honeybees.

### 4. Next Steps

We aim to run similar tests with a larger number of bees so we can compare our scenting data from a variety of trials in



(g) Two honeybees time\_series\_bar



(h) Three honeybees time\_series\_bar

order to better understand the behavior of individuals in the presence of the swarm.

Along with this, we hope to improve the tracking algorithm to require less human interaction yet still result in the same high tracking accuracies.

Finally we will analyze the resulting distributions and attempt to understand the correlation between the data for each of our trials and work towards understanding how the variations between tests relate to the honeybee swarm dynamics.

### References

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